DEPT. OF APPLIED MATHEMATICS

SAME STATS DIFFERENT GRAPHS

STATISTICAL VALIDATION AND GEOMETRIC MORPHING VIA SIMULATED ANNEALING

TEAM

ANIRUDHHA PANDEY | 26-27-17 | M.TECH DATA SCIENCE

SOURABH DANG | 26-27-18 | M.TECH DATA SCIENCE

RITIK JOSHI | 26-27-16 | M.TECH DATA SCIENCE

ASTHA DESHMUKH | 26-14-06 | M.TECH MODELING & SIMULATION

PROBLEM STATEMENT

- Datasets are routinely reduced to mean, variance, correlation, and a regression line
- This abstraction is mathematically incomplete
- Very different distributions can produce identical summary statistics
- Without visualization, skewed or clustered data can pass as normal, leading to flawed models and corrupted data pipelines

Key message: Never trust the numbers without plotting the data.

OBJECTIVES

- Pure NumPy statistical engine: mean, variance, Pearson r, and OLS with no black-box libraries
- Anscombe's Quartet validation: prove four distinct datasets share identical statistics
- Datasaurus Dozen validation: extend the proof to 13 complex shapes
- Stochastic shape morphing: use Simulated Annealing to turn one point cloud into another while keeping the statistics locked

LITERATURE FOUNDATION

- **Anscombe (1973):** numbers alone are not enough, so graph the data. His datasets calibrated our engine.
- **Matejka & Fitzmaurice (2017)** showed that point clouds can be steered into target shapes via Simulated Annealing. We adapted their acceptance rule:

$$P(\text{Accept}) = \exp(-\Delta\text{Cost} / T)$$

PIPELINE & ETL (PHASE I)

FLOW: ETL → ANSCOMBE → DATASAURUS → SIMULATED ANNEALING

- Input: anscombe.csv, datasaurus.csv
- Strip nulls, resolve coordinate overlaps, cast to float64
- Output: anscombe_clean.csv, datasaurus_clean.csv
- Why: prevents floating-point errors from causing false rejections later

DATA SANITIZATION

- Why it matters: raw data must be clean before statistical processing, so the Simulated Annealing engine never sees corrupt values.

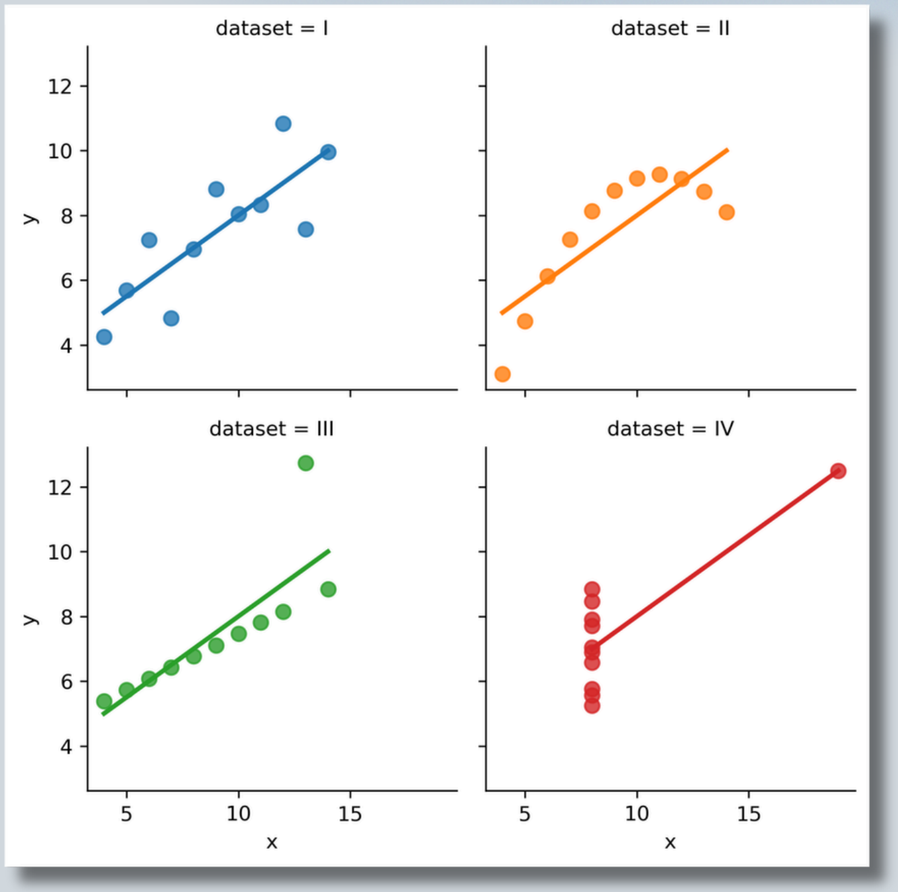
  Extract → Transform → Load
- Extract: ingest the raw source files anscombe.csv and datasaurus.csv
- Transform:Strip null values
- Resolve coordinate overlaps
- Cast all spatial data to float64 precision
- Load: write new sanitized files, anscombe_clean.csv and datasaurus_clean.csv
- Impact: the annealing iterations run on pristine data, so floating-point rounding errors can’t trigger false rejections by the Gatekeeper.

STATISTICAL ENGINE (PURE NUMPY)

- Mean: $\bar{x} = (1/n)\ \Sigma x_i$
- Variance (Bessel's correction): $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$
- Pearson $r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$
- OLS: $\beta = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$, $\alpha = \bar{y} - \beta\bar{x}$

ANSCOMBE'S QUARTET RESULTS (PHASE II)

Metric	Value (all 4 datasets)
X mean	9.00
Y mean	7.50
X variance	11.00
Y variance	4.12
Correlation	0.816



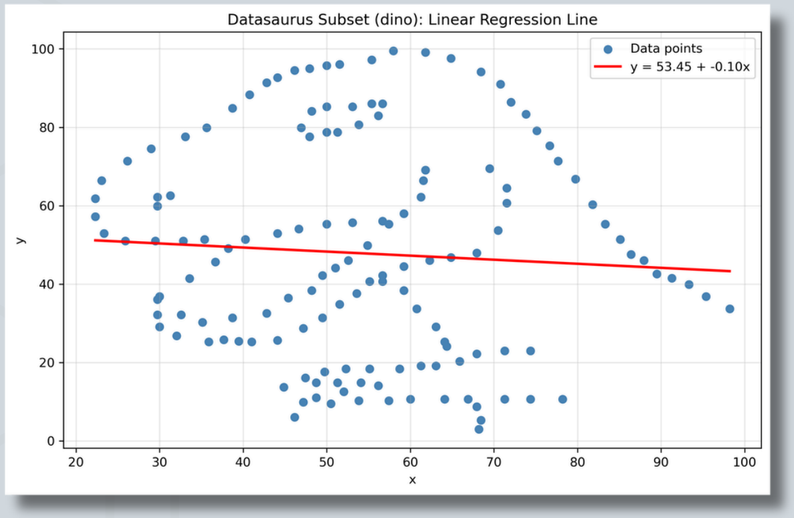
- What we did
  - Proved that four very different datasets share identical statistics.
- How we did it
  - Loaded anscombe_clean.csv (four separate coordinate arrays)
  - Built a pure NumPy engine with no statistical libraries:
  - Mean
  - Variance (Bessel's correction, n − 1)
  - Pearson r
  - OLS regression
- Calibrated the engine against Anscombe's 1973 published values

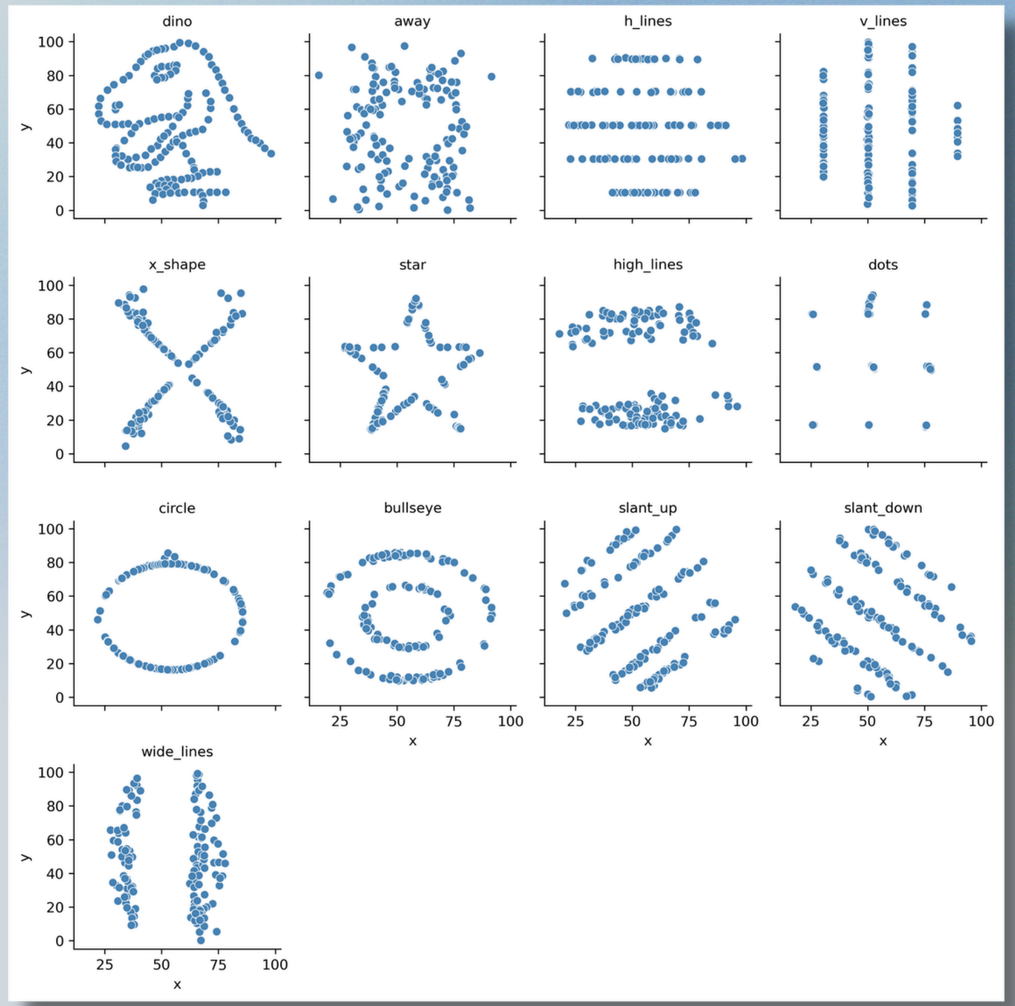
Result (identical for all four datasets)

- X mean 9.00 | Y mean 7.50 | X variance 11.00 | Y variance 4.12 | r = 0.816 | y = 3.00 + 0.50x
- Takeaway: the plots are linear, curved, and outlier-driven, yet the numbers match.

DATASAURUS DOZEN (PHASE III)

- 13 distinct shapes (dino, star, circle, bullseye, x_shape, lines, etc.), same statistics
- Dino baseline:
  - Mean X = 54.26, Y = 47.83
  - Variance X = 281.07, Y = 725.52
  - r = −0.0645
  - OLS: y = 53.45 − 0.10x





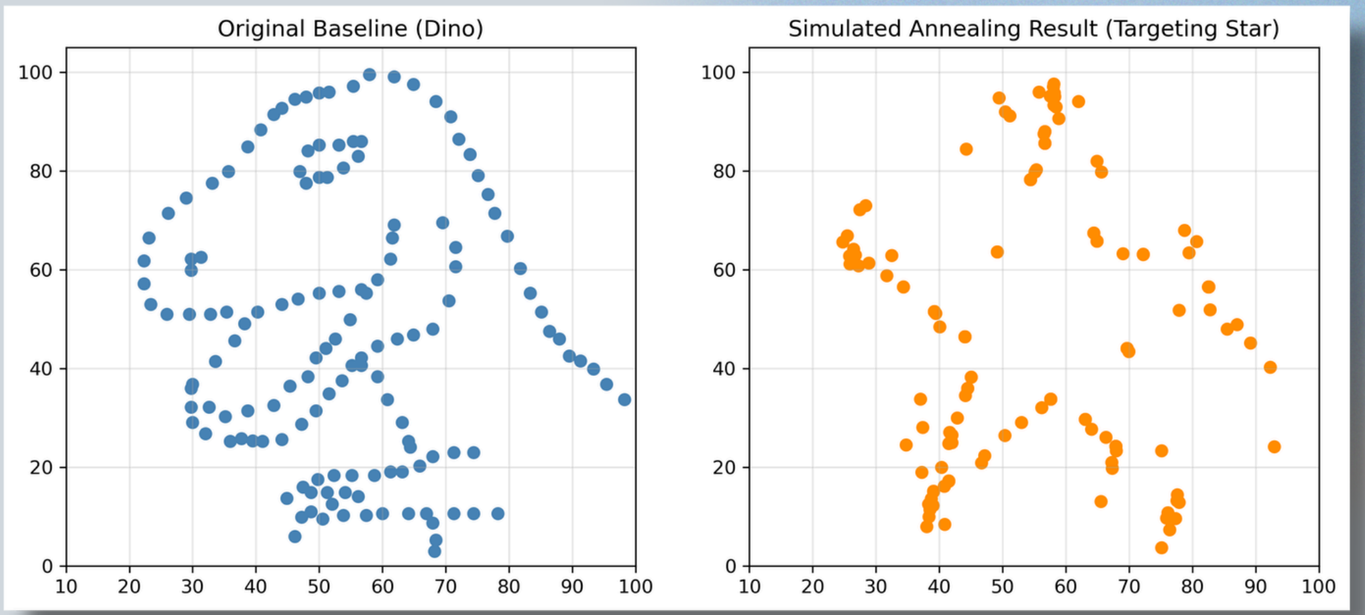
- What we did
  - Scaled the proof from 4 datasets to 13 complex shapes (dino, star, circle, bullseye, x_shape, lines, etc.).
- How we did it
  - Loaded datasaurus_clean.csv
  - Split it into subsets with Pandas groupby
  - Ran the same NumPy engine on every subset
  - Locked the dino subset's statistics as the baseline constraints for Phase IV

Result (dino baseline)

- Mean X = 54.26, Y = 47.83 | Variance X = 281.07, Y = 725.52 | r = −0.0645 | y = 53.45 − 0.10x
- Takeaway: all 13 shapes return the same statistics, so summary numbers can hide extreme spatial differences.

SIMULATED ANNEALING ENGINE (PHASE IV)

- Goal: morph dino → star
- Move: perturb a random point with Gaussian noise $N[0, \sigma^2]$
- Cost: Euclidean distance to the star target
- Gatekeeper: reject any move where mean, variance, Pearson r, or regression drifts more than 0.1 from baseline
- Acceptance: better moves always accepted; worse moves accepted with probability $\exp[-\Delta Cost / T]$
- Parameters: $T_0 = 100$, cooling 0.9995, 150,000 iterations



- What we did
  - Morphed the dino point cloud into a star while keeping its statistics unchanged.
- How we did it
  - Perturb: shift a random point with Gaussian noise $N(0, \sigma^2)$
  - Gatekeeper: recompute mean, variance, r, and regression; reject the move if any drifts more than 0.1 from baseline
  - Cost: Euclidean distance to the star target
  - Accept: better moves always; worse moves with probability $\exp(-\Delta\text{Cost} / T)$
  - Parameters: $T_0 = 100$, cooling rate 0.9995, 150,000 iterations

Result

- Cost fell from 1103.80 to 216.86, an 80.35% reduction in spatial divergence.
- Takeaway: data can be reshaped on purpose without triggering any statistical alarm.

CONCLUSION & REFERENCES

- Summary statistics are an incomplete description of data
- Different geometries can share identical mean, variance, correlation, and regression
- Data can be deliberately shaped without triggering statistical alarms
- Takeaway: always visualize before modeling

References

1. Anscombe, F. J. (1973). Graphs in Statistical Analysis. The American Statistician, 27(1), 17–21.
2. Matejka, J., & Fitzmaurice, G. (2017). Same Stats, Different Graphs. CHI 2017, 1290–1294.

Thank You